

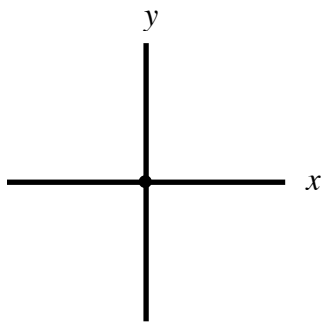
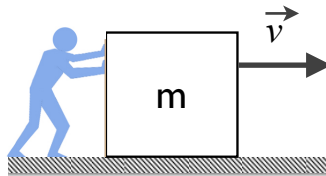
Friction

Concept: Friction is a force exerted by surfaces when they slide (kinetic friction f_k) or tend to slide (static friction f_s) over one another. Friction between sliding solids depends on the type of surfaces (via the coefficient of friction μ) and the magnitude of the normal force n .

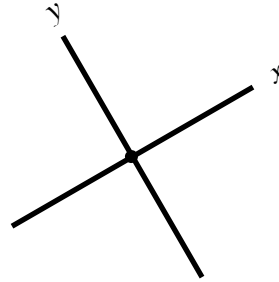
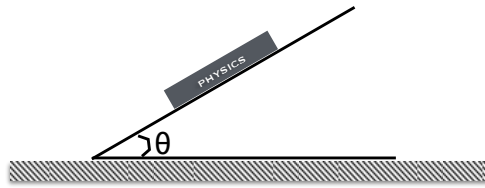
$$f_k = \mu_k n$$

$$f_s \leq f_{s \max} = \mu_s n$$

Problem 1: A student pushes a 50 kg wood box across a wood floor. It accelerates at a constant rate of 1.0 m/s^2 . The coefficient of friction between the surfaces is 0.2. How much force does the student exert on the box?



Problem 2: A student tilts her desk to make her physics book slide to the floor. The book is about to start sliding at an angle θ . Determine the coefficient of static friction μ_s between the book and the desk.



Going Sledding: Your little sister wants you to give her a ride on her sled. On level ground, what is the easiest way to accomplish this, for the case where there's friction between the sled and the ground?

- pushing her from behind
- pulling her from the front
- both are equivalent
- it is impossible to move the sled

